1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
d2_96712_Devanand> CREATE PROCEDURE rectangle_calc_param(IN L INT, IN W INT)
    -> BEGIN
    ->     DECLARE perimeter INT;
    ->     DECLARE area INT;
    ->
    ->     SET perimeter = 2 * (L + W);
    ->     SET area = L * W;
    ->
    ->     SELECT perimeter AS Perimeter, area AS Area;
    -> END //
Query OK, 0 rows affected (0.237 sec)

d2_96712_Devanand>
d2_96712_Devanand> DELIMITER ;
d2_96712_Devanand> CALL rectangle_calc();
ERROR 1305 (42000): PROCEDURE hr.rectangle_calc does not exist
d2_96712_Devanand> CALL rectangle_calc_param(10, 5);
+-----------+------+
| Perimeter | Area |
+-----------+------+
|        30 |   50 |
+-----------+------+
1 row in set (0.035 sec)

Query OK, 0 rows affected (0.113 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
d2_96712_Devanand> CREATE TEMPORARY TABLE temp_numbers (
    ->     num INT,
    ->     square_val INT,
    ->     cube_val INT
    -> );^C
d2_96712_Devanand> CREATE TEMPORARY TABLE temp_numbers (
    ->     num INT,
    ->     square_val INT,
    ->     cube_val INT
    -> );
Query OK, 0 rows affected (0.547 sec)

d2_96712_Devanand> DELIMITER //
d2_96712_Devanand>
d2_96712_Devanand> CREATE PROCEDURE calc_num_values()
    -> BEGIN
    ->     DECLARE num INT DEFAULT 5;
    ->
    ->     INSERT INTO temp_numbers (num, square_val, cube_val)
    ->     VALUES (
    ->         num,
    ->         num * num,
    ->         num * num * num
    ->     );
    ->
    ->     SELECT * FROM temp_numbers;
    -> END //
Query OK, 0 rows affected (0.232 sec)

d2_96712_Devanand>
d2_96712_Devanand> DELIMITER ;
d2_96712_Devanand> CALL calc_num_values();
+------+------------+----------+
| num  | square_val | cube_val |
+------+------------+----------+
|    5 |         25 |      125 |
+------+------------+----------+
1 row in set (0.166 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32

```
d2_96712_Devanand> CREATE PROCEDURE temp_convert(
    ->     IN choice INT,     -- 1 for F→C, 2 for C→F
    ->     IN temp_val DOUBLE
    -> )
    -> BEGIN
    ->     DECLARE result DOUBLE;
    ->
    ->     IF choice = 1 THEN
    ->         -- Fahrenheit to Celsius
    ->         SET result = (temp_val - 32) * 5 / 9;
    ->         SELECT temp_val AS Fahrenheit, result AS Celsius;
    ->
    ->     ELSEIF choice = 2 THEN
    ->         -- Celsius to Fahrenheit
    ->         SET result = (9 / 5) * temp_val + 32;
    ->         SELECT temp_val AS Celsius, result AS Fahrenheit;
    ->
    ->     ELSE
    ->         SELECT 'Invalid Choice (Use 1 or 2)' AS Message;
    ->     END IF;
    ->
    -> END //
Query OK, 0 rows affected (0.138 sec)

d2_96712_Devanand>
d2_96712_Devanand> DELIMITER ;
d2_96712_Devanand> CALL temp_convert(1, 98.6);
+------------+---------+
| Fahrenheit | Celsius |
+------------+---------+
|       98.6 |      37 |
+------------+---------+
1 row in set (0.159 sec)
```

4.Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
d2_96712_Devanand> DELIMITER //
d2_96712_Devanand>
d2_96712_Devanand> CREATE PROCEDURE convert_inches(IN total_inches INT)
    -> BEGIN
    ->     DECLARE yards INT;
    ->     DECLARE feet INT;
    ->     DECLARE inches INT;
    ->
    ->     -- Calculate yards
    ->     SET yards = total_inches DIV 36;
    ->
    ->     -- Remaining inches after yards
    ->     SET total_inches = total_inches MOD 36;
    ->
    ->     -- Calculate feet
    ->     SET feet = total_inches DIV 12;
    ->
    ->     -- Remaining inches
    ->     SET inches = total_inches MOD 12;
    ->
    ->     SELECT yards AS Yards,
    ->            feet AS Feet,
    ->            inches AS Inches;
    -> END //
Query OK, 0 rows affected (0.235 sec)

d2_96712_Devanand>
d2_96712_Devanand> DELIMITER ;
d2_96712_Devanand> CALL convert_inches(124);
+-------+------+--------+
| Yards | Feet | Inches |
+-------+------+--------+
|     3 |    1 |      4 |
+-------+------+--------+
1 row in set (0.119 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
d2_96712_Devanand> DELIMITER //
d2_96712_Devanand>
d2_96712_Devanand> CREATE PROCEDURE check_product(
    ->     IN num1 DOUBLE,
    ->     IN num2 DOUBLE
    -> )
    -> BEGIN
    ->     DECLARE product DOUBLE;
    ->
    ->     SET product = num1 * num2;
    ->
    ->     IF product >= 100 THEN
    ->         SELECT num1 AS Number1,
    ->                num2 AS Number2,
    ->                product AS Product,
    ->                'Product is greater than or equal to 100' AS Result;
    ->     ELSE
    ->         SELECT num1 AS Number1,
    ->                num2 AS Number2,
    ->                product AS Product,
    ->                'Product is less than 100' AS Result;
    ->     END IF;
    ->
    -> END //
Query OK, 0 rows affected (0.234 sec)

d2_96712_Devanand>
d2_96712_Devanand> DELIMITER ;
d2_96712_Devanand> CALL check_product(10, 12);
+---------+---------+---------+----------------------------------------+
| Number1 | Number2 | Product | Result                                 |
+---------+---------+---------+----------------------------------------+
|      10 |      12 |     120 | Product is greater than or equal to 100 |
+---------+---------+---------+----------------------------------------+
1 row in set (0.126 sec)
```